Solutions

1

The values are $p(0) = 1$ and $p(1) = 1 + 1 + 1 = 1 \pmod 2$. A reducible degree-3 polynomial must have a linear factor, so the lack of roots proves irreducibility. The eight quotient representatives are

$$0, 1, x, x + 1, x^2, x^2 + 1, x^2 + x, x^2 + x + 1.$$

All have degree below 3, and division uniqueness makes their classes distinct.

Euclidean division gives

$$p = x(x^2) + (x + 1), \qquad x^2 = (x + 1)(x + 1) + 1.$$

Back-substitution in characteristic 2 gives

$$\begin{aligned} 1 &= x^2 + (x + 1)(p + x \cdot x^2) \\ &= (x^2 + x + 1)x^2 + (x + 1)p. \end{aligned}$$

Therefore $[x^2]^{-1} = [x^2 + x + 1]$. Multiplication gives $x^4 + x^3 + x^2$. Reduce with $x^3 = x + 1, x^4 = x^2 + x$:

$$(x^2 + x) + (x + 1) + x^2 = 1.$$

All three calculations use the same quotient relation.

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2

Since $(\alpha + 1)^2 = \alpha^2 + 1 = (\alpha + 1) + 1 = \alpha$,

$$(\alpha + 1)^3 = \alpha(\alpha + 1) = 1, \qquad (\alpha + 1)^2 + \alpha = \alpha + \alpha = 0.$$

Multiplying back verifies every nonzero inverse:

$$1^{-1} = 1, \quad \alpha^{-1} = \alpha + 1, \quad (\alpha + 1)^{-1} = \alpha.$$

Every nonzero z has $z + z = 0$, so its additive order is 2. In $\mathbb{Z}_4$, the additive order of 1 is 4. Additive-group isomorphisms preserve element orders. Thus the additive groups, and therefore the rings, are not isomorphic.

3

With rows $x = 0, 1, 2$ and columns $y = 0, 1, 2$, the squares are

$$L_1 = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 0 \\ 2 & 0 & 1 \end{bmatrix}, \qquad L_2 = \begin{bmatrix} 0 & 1 & 2 \\ 2 & 0 & 1 \\ 1 & 2 & 0 \end{bmatrix}.$$

Every row and column lists 0, 1, 2 once, so both are Latin. The first row and first column of L_1 are 0, 1, 2, so it is reduced. The first row of L_2 is normalized, but its first column is 0, 2, 1. In this fixed ordering, it is not reduced.

Solutions (continued)

For a symbol pair (u, v), solving $x + y = u, 2x + y = v$ gives the unique solution $x = v - u, y = 2u - v$ in $\mathbb{F}_3$. Hence the squares are orthogonal. Swapping rows of only the second square to make it reduced changes cell pairings. Orthogonality cannot then be assumed to remain valid.

4

The point set is $\mathbb{F}_3^2$, so $v = 9$. Each slope $m = 0, 1, 2$ has three intercepts, and there are three vertical lines. Thus $b = 3 \cdot 3 + 3 = 12$. Each line allows one freely chosen coordinate, giving $k = 3$. One line in each of the four directions passes through each point, so $r = 4$.

The two given points have different x-coordinates. Compute

$$m = (1 - 2)/(2 - 1) = 2, \qquad b = 2 - 2 \cdot 1 = 0 \pmod 3.$$

Their line is $y = 2x$, with points $(0, 0), (1, 2), (2, 1)$.

A parallel through $(1, 1)$ must satisfy $1 = 2 \cdot 1 + b$, so $b = 2$. The line $y = 2x + 2$ contains $(0, 2), (1, 1), (2, 0)$. An intersection would satisfy $2x = 2x + 2$, forcing $0 = 2$, a contradiction. The lines are disjoint.

5

In a $2 - (7, 3, 1)$ design, every block contains $\binom{3}{2} = 3$ pairs. Each of the $\binom{7}{2} = 21$ pairs occurs once. Hence $3b = 21$, giving $b = 7$. Fix a point P. Each block through P supplies two other points, and each of the six other points occurs with P once. Thus $2r = 6$, giving $r = 3$. The count $vr = bk = 21$ agrees.

For $2 - (16, 6, 1)$, necessary counting gives

$$r = 15/5 = 3, \qquad b = 16 \cdot 3/6 = 8.$$

Both values are integral. However, Fisher's inequality for a nondegenerate BIBD requires $b \geq v$, whereas $8 < 16$. Its hypotheses $2 \leq k < v$ and $\lambda \geq 1$ hold here, so the design is impossible.

6

Points are one-dimensional subspaces. Distinct points P, Q intersect only at 0, so their span has dimension 2. Any two-dimensional subspace containing both must be this span, proving uniqueness of their line.

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Solutions (continued)

Distinct two-dimensional subspaces $U, V \subset F_q^3$ satisfy $\dim(U \cap V) \geq 2 + 2 - 3 = 1$. An intersection of dimension 2 would make them equal. Their intersection is therefore one-dimensional, giving their unique projective point.

There are $q^3 - 1$ nonzero vectors. Each point groups the $q - 1$ nonzero scalar multiples of a representative, so

$$v = (q^3 - 1)/(q - 1) = q^2 + q + 1.$$

A two-dimensional line has $q^2 - 1$ nonzero vectors. Grouping these by scalar multiples gives $(q^2 - 1)/(q - 1) = q + 1$ points per line.

Two-dimensional subspaces containing a fixed one-dimensional P correspond to one-dimensional subspaces of the two-dimensional quotient F_q^3/P. Thus

$$r = (q^2 - 1)/(q - 1) = q + 1.$$

Here the groups consist of nonzero quotient vectors, rather than all original three-dimensional vectors.

Lines are also kernels of nonzero linear functionals. Functionals have the same kernel exactly when they differ by a nonzero scalar. Grouping the $q^3 - 1$ nonzero functionals into groups of $q - 1$ gives $b = q^2 + q + 1$. These are derivations in the subspace model. Abstract planes instead take the two uniqueness statements as axioms.

7

The values of $x^2 + 1$ in $\mathbb{F}_3$ are 1, 2, 2, so it is irreducible and the quotient is a field. Since $\alpha^2 = 2$, $\alpha^4 = 1$, and $\alpha^2 \neq 1$, we have $o(\alpha) = 4$, smaller than the multiplicative group order 8.

Set $\beta = 1 + \alpha$. Repeated multiplication and reduction give

i	0	1	2	3	4	5	6	7
β^i	1	$1 + \alpha$	2α	$1 + 2\alpha$	2	$2 + 2\alpha$	α	$2 + \alpha$

For example, $\beta^3 = 2\alpha(1 + \alpha) = 2\alpha + 4 = 1 + 2\alpha$, $\beta^4 = (2\alpha)^2 = 2$, and $\beta^8 = 1$. Lagrange's theorem says the order divides 8. The inequality $\beta^4 \neq 1$ excludes orders 1, 2, 4, so the order is 8.

The inverse is $\beta^7 = 2 + \alpha$, since

$$(1 + \alpha)(2 + \alpha) = 2 + 3\alpha + \alpha^2 = 2 + 2 = 1 \pmod 3.$$

Solutions (continued)

8

Since $f(0) = f(1) = 1$, there is no linear factor. If a degree-4 polynomial with no linear factor is reducible, it must be the product of two monic irreducible quadratics.

A monic quadratic $x^2 + ax + b$ over $\mathbb{F}_2$ with no roots must have $b = 1$ to exclude root 0. To exclude root 1, it must have $1 + a + b = a = 1$. Thus the only such quadratic is $h = x^2 + x + 1$. But

$$h^2 = x^4 + x^2 + 1 \neq x^4 + x + 1.$$

Therefore f is irreducible, and the quotient field has 16 elements.

Let $\gamma = [x]$. Then $\gamma^4 = \gamma + 1$ and $\gamma^5 = \gamma^2 + \gamma$. Its order divides the nonzero group size 15, whose proper divisors are 1, 3, 5. Uniqueness of representatives of degree below 4 gives

$$\gamma \neq 1, \qquad \gamma^3 \neq 1, \qquad \gamma^5 = \gamma^2 + \gamma \neq 1.$$

The order must therefore be 15.

9

In both arrays, fixing x makes translation in y. Fixing y makes multiplication in x by 1 or 3, both units of $\mathbb{Z}_4$. Thus each array is Latin.

However, the different cells $(x, y) = (0, 0)$ and $(2, 2)$ both give the ordered pair $(0, 0)$, since

$$(L_1(2, 2), L_3(2, 2)) = (4, 8) \equiv (0, 0) \pmod 4.$$

The squares are not orthogonal. Solving the simultaneous equations in the field proof requires division by $1 - 3 = 2 \pmod 4$, but 2 has no inverse. This is the failed step.

10

The three squares are $L_1, L_\alpha, L_{\alpha+1}$. Their first columns are $ax + 0$, with rows ordered $0, 1, \alpha, \alpha + 1$:

a	First column
1	$0, 1, \alpha, \alpha + 1$
α	$0, \alpha, \alpha + 1, 1$
$\alpha + 1$	$0, \alpha + 1, 1, \alpha$

Solutions (continued)

Each nonzero coefficient is invertible, so all rows and columns are permutations. For distinct a, b, every (u, v) has the unique preimage

$$x = (a - b)^{-1}(u - v), \qquad y = u - ax.$$

Thus the three squares are pairwise orthogonal.

For $a = \alpha, b = \alpha + 1, u = 1, v = \alpha$, characteristic 2 gives

$$a - b = 1, \qquad u - v = \alpha + 1, \qquad x = \alpha + 1, \qquad y = 1 - \alpha(\alpha + 1) = 0.$$

Direct substitution gives $L_\alpha(x, 0) = 1$ and $L_{\alpha+1}(x, 0) = (\alpha + 1)^2 = \alpha$, as required.

11

The blocks containing each point are

1	123, 145, 167
2	123, 246, 257
3	123, 347, 356
4	145, 246, 347
5	145, 257, 356
6	167, 246, 356
7	167, 257, 347

Every point has replication 3. Replace labels with their nonzero three-bit vectors. The XOR of every block is 000. For distinct u, v, the unique third point is $u + v$, so no other third point can complete the same pair to a block. Every pair is covered exactly once. The seven blocks each contain three pairs, giving 21, equal to the total number of pairs.

After deleting line 123, the three parallel classes are

$$\{\{4, 5\}, \{6, 7\}\}, \qquad \{\{4, 6\}, \{5, 7\}\}, \qquad \{\{4, 7\}, \{5, 6\}\}.$$

They are labeled by the deleted points 1, 2, 3 respectively.

Chapter 16 orders the Fano labels of the columns of H as $6, 5, 3, 7, 4, 2, 1$. Line 145 therefore gives positions 7, 5, 2 and the word 0100101. Those columns satisfy $101 + 100 + 001 = 000$, so this is a codeword. Fano labels 1, 4, 5 are not themselves the column positions of H.

12

Each row contains three ones, and two distinct rows share one block. Hence

$$MM^T = 2I + J.$$

Solutions (continued)

On the all-ones vector, J has eigenvalue 7, so $2I + J$ has eigenvalue 9. On the six-dimensional subspace of vectors with coordinate sum zero, its eigenvalue is 2. Therefore

$$(\det M)^2 = \det(MM^T) = 9 \cdot 2^6 = 576, \qquad |\det M| = 24.$$

Using $J^2 = 7J$, direct multiplication verifies

$$(2I + J)^{-1} = \tfrac{1}{2}I - \tfrac{1}{18}J.$$

Every block has size 3, so $M^T J = 3J$, and

$$M^{-1} = M^T(MM^T)^{-1} = \tfrac{1}{2}M^T - \tfrac{1}{6}J.$$

Check by multiplying back:

$$M(\tfrac{1}{2}M^T - \tfrac{1}{6}J) = \tfrac{1}{2}(2I + J) - \tfrac{1}{6}(3J) = I.$$

Over $\mathbb{F}_2$, the determinant is the integer determinant reduced modulo 2. Since $24 \equiv 0$, the matrix is singular. Positive definiteness and the fractional inverse formula apply over the reals. In characteristic 2, $1/2$ does not exist, so those arguments cannot be copied.

13

Let $e_1, \dots, e_d$ be a basis of E over $\mathbb{F}_p$, and $u_1, \dots, u_m$ a basis of K over E. The latter is finite because K is finite. Write any $x \in K$ as $\sum_j a_j u_j$, then write $a_j = \sum_i c_{ij} e_i$. Thus the products $\{e_i u_j\}$ span K over $\mathbb{F}_p$.

If

$$\sum_j \left(\sum_i c_{ij} e_i \right) u_j = 0,$$

the E-linear independence of the u_j gives $\sum_i c_{ij} e_i = 0$ for every j. The $\mathbb{F}_p$-linear independence of the e_i then gives every $c_{ij} = 0$. These dm products form a basis. Hence $n = dm$, proving $d \mid n$.

An embedding of $GF(4)$ into $GF(8)$ would have a subfield as its image. Their dimensions over $\mathbb{F}_2$ are 2 and 3, but $2 \nmid 3$. No such embedding exists. The inequality $4 < 8$ between set sizes does not suffice.

14

The group $K^\times$ is cyclic of order 15. Choose a generator g. A nonzero element $a = g^j$ satisfies

$$a^4 = a \iff a^3 = 1 \iff 15 \mid 3j.$$

Solutions (continued)

For $0 \leq j < 15$, the solutions are $j = 0, 5, 10$. These are three nonzero elements; adding 0 gives exactly four elements in S.

The characteristic is 2, so $(a+b)^4 = a^4 + b^4$. For $a, b \in S$, this gives $(a+b)^4 = a + b$. Also $(ab)^4 = a^4 b^4 = ab$. Additive inverses equal the original elements and remain in S. For $a \neq 0$,

$$(a^{-1})^4 = (a^4)^{-1} = a^{-1}.$$

The set contains 0 and 1. It is therefore a four-element subfield, isomorphic to $GF(4)$.

The root bound alone would give at most four roots. The cyclic-group count proves there are exactly four.

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Solutions (continued)

Verification code appendix: polynomial_arithmetic.py

```
"""Polynomial arithmetic over F_p, p prime. Coefficients are low degree first."""


def trim(a, p):
    result = [x % p for x in a]
    while result and result[-1] == 0:
        result.pop()
    return result


def add(a, b, p):
    return trim([(a[i] if i < len(a) else 0) +
                 (b[i] if i < len(b) else 0)
                 for i in range(max(len(a), len(b)))], p)


def neg(a, p):
    return trim([-x for x in a], p)


def multiply(a, b, p):
    a, b = trim(a, p), trim(b, p)
    if not a or not b:
        return []
    result = [0] * (len(a) + len(b) - 1)
    for i, x in enumerate(a):
        for j, y in enumerate(b):
            result[i + j] += x * y
    return trim(result, p)


def divide(a, b, p):
    """Return quotient/remainder; [] is the zero polynomial, degree -infinity."""
    remainder, divisor = trim(a, p), trim(b, p)
    if not divisor:
        raise ZeroDivisionError('zero polynomial divisor')
    quotient = [0] * max(0, len(remainder) - len(divisor) + 1)
    while remainder and len(remainder) >= len(divisor):
        shift = len(remainder) - len(divisor)
        coefficient = remainder[-1] * pow(divisor[-1], -1, p) % p
        quotient[shift] += coefficient
```

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Solutions (continued)

```
        subtract = [0] * shift + [coefficient * x for x in divisor]
        remainder = add(remainder, neg(subtract, p), p)
    return trim(quotient, p), remainder


def extended_gcd(a, b, p):
    """Return monic gcd d and coefficients s,t satisfying d=s*a+t*b."""
    r0, r1 = trim(a, p), trim(b, p)
    s0, s1, t0, t1 = [1], [], [], [1]
    while r1:
        q, remainder = divide(r0, r1, p)
        r0, r1 = r1, remainder
        s0, s1 = s1, add(s0, neg(multiply(q, s1, p), p), p)
        t0, t1 = t1, add(t0, neg(multiply(q, t1, p), p), p)
    if not r0:
        return [], [], []
    scale = pow(r0[-1], -1, p)
    return tuple(trim([scale * c for c in poly], p) for poly in (r0, s0, t0))


def inverse(a, modulus, p):
    """Return inverse in F_p[x]/(modulus); fail if the class is not a unit."""
    modulus = trim(modulus, p)
    if len(modulus) < 2:
        raise ValueError('modulus must have positive degree')
    d, s, _ = extended_gcd(a, modulus, p)
    if d != [1]:
        raise ValueError('polynomial class is not a unit')
    return divide(s, modulus, p)[1]


if __name__ == '__main__':
    # x^3+2x^2+4 divided by x^2+2 in F5.
    assert divide([4, 0, 2, 1], [2, 0, 1], 5) == ([2, 1], [0, 3])
    # In F2[x]/(x^3+x+1), inverse of x^2 is x^2+x+1.
    assert inverse([0, 0, 1], [1, 1, 0, 1], 2) == [1, 1, 1]
    # In F3[x]/(x^2+1), inverse of 1+x is 2+x.
    assert inverse([1, 1], [1, 0, 1], 3) == [2, 1]
    print('polynomial division and quotient inverses: PASS')
```

References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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